Year 2 Statistics and Mechanics

Chapter 2

Conditional probability

Practice Booklet

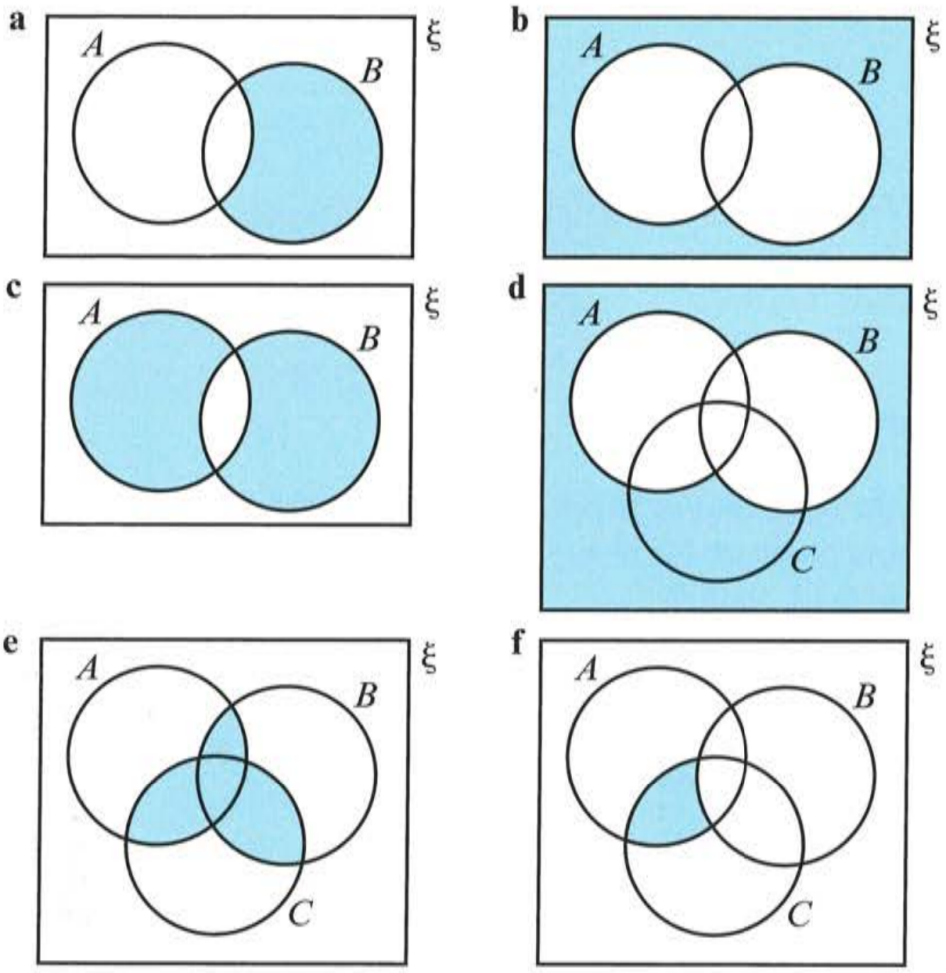
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2.1 Set notation

Practice Book, printed page 9.

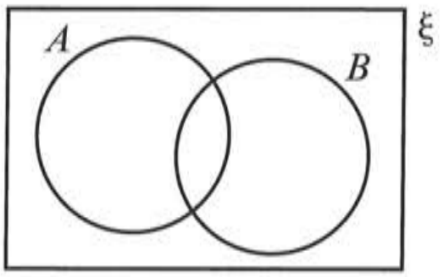
1. Use set notation to describe the shaded area in each of these Venn diagrams:



Hint

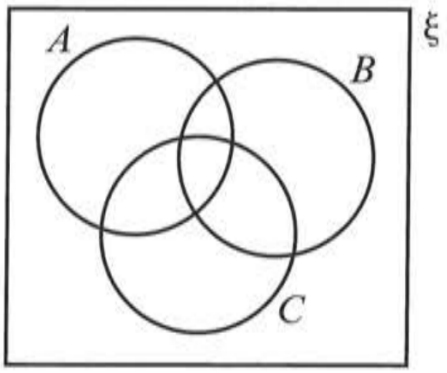
The intersection of A and B is written as $A \cap B$.
The union of A and B is written as $A \cup B$.
The complement of A is written as A'.

2. On separate copies of this Venn diagram, shade:



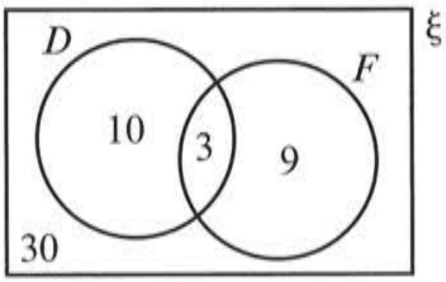
a. $A \cap B'$
b. $A' \cup B'$
c. $(A \cap B) \cup (A' \cap B')$

3. On separate copies of this Venn diagram, shade:



a. $A \cup (B \cap C)$
b. $(A \cup B) \cap C'$
c. $(A \cup (B \cap C'))'$

4. A card is chosen at random from a pack of 52 playing cards. D is the event 'the card chosen is a diamond' and F is the event 'the card chosen is a Jack, Queen or King'.
The Venn diagram shows the number of outcomes for each event.



Find:

a. $\mathrm{P}(F)$
b. $\mathrm{P}(D)$
c. $\mathrm{P}(D \cap F)$
d. $\mathrm{P}(D \cup F)$
e. $\mathrm{P}(F')$
f. $\mathrm{P}(D' \cap F)$

5. A and B are two events. $\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.3$ and $\mathrm{P}(A \cap B) = 0.2$.
Find:

a. $\mathrm{P}(A \cup B)$

b. $\mathrm{P}(A')$

c. $\mathrm{P}(A' \cap B)$

d. $\mathrm{P}(A' \cap B')$

Hint

Draw a Venn diagram showing the probabilities for each event. Remember that all the probabilities on your Venn diagram must add up to 1.

6. An insurance company surveys the behaviour of people who visit its website on one day. The results are that 40% of visitors purchase car insurance, 25% of visitors purchase home insurance and 45% of visitors purchase neither.

a. Draw a Venn diagram to illustrate these percentages.

A visitor is chosen at random from those surveyed.

b. Find the probability that the visitor purchased:

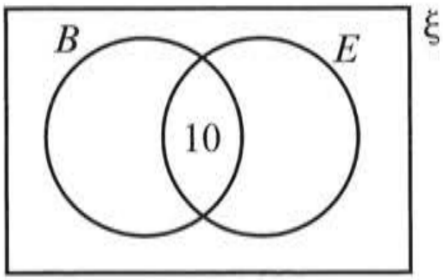
i car and home insurance

ii car insurance but not home insurance

iii home insurance but not car insurance.

7. P A roulette wheel has 37 slots labelled 0 to 36. Slot 0 is green.
Half of the slots numbered 1 to 36 are red and half are black.
There are 10 black numbers that are even.
A ball is spun and lands in one of the slots.
B is the event that the ball lands in a black slot.
E is the event that the ball lands on an even number.

a. Copy and complete the Venn diagram showing the numbers of possible outcomes.



b. Find:

i $\mathrm{P}(B \cup E)$

ii $\mathrm{P}(B' \cap E)$

iii $\mathrm{P}((B \cap E)')$

8. E/P Twenty mathematicians from a university are attending a conference.
The tables show which of three seminars, A, B and C, each mathematician will be attending.

	A	B	C		A	B	C
Agnesi	✓			Kovalevskaya			
Birman			✓	Lovelace			✓
Cartwright		✓		Mirzakhani	✓		
Darden	✓	✓		Noether		✓	
Easley	✓			Owens		✓	
Faddeeva		✓	✓	Paulson	✓		
Germain		✓		Quinn	✓	✓	
Hypatia	✓			Robinson		✓	✓
Inniss			✓	Somerville	✓		
Johnson	✓			Tinney			

A mathematician is chosen at random from the twenty attendees.
A, B and C are the events that the mathematician attends each of the three seminars.

a. Draw a Venn diagram showing the probabilities of each event. (3 marks)

b. Show that:

i A and C are mutually exclusive

ii B and C are independent.

(3 marks)

c. Find:

i $\mathrm{P}(A' \cap C)$

ii $\mathrm{P}(A \cup (B' \cup C))$

iii $\mathrm{P}((A \cup C)' \cap B)$

(3 marks)

2.2 Conditional probability

Practice Book, printed page 11.

1. The two-way table shows the languages studied by 80 students in a sixth-form.

	French	**Spanish**	**Total**
Male	24	18	42
Female	16	22	38
Total	40	40	80

Find:

a. P(Male)

b. P(French | Male)

c. P(Male | Spanish)

d. P(Spanish | Female)

Hint

The probability that a student is studying French given that they are male is written as P(French | Male). There are 42 males in total, and 24 of them are studying French.

2. A green spinner and a yellow spinner each have four equally likely outcomes, numbered 1 to 4. The two spinners are spun at the same time, and the product of the numbers shown, X, is recorded.

a. Draw a sample space diagram for X.

b. Find:

i $\mathrm{P}(X = 8)$

ii $\mathrm{P}(X = 6 \mid \text{Green spinner is 2})$

iii $\mathrm{P}(\text{Yellow spinner is 2} \mid X = 4)$

Hint

Draw your sample space diagram as a two-way table with 4 row headings and 4 column headings.

3. Two fair six-sided dice are thrown and the larger number, Y, is recorded.

a. Draw a sample space diagram to illustrate the possible outcomes.

b. Find $\mathrm{P}(Y = 4)$.

c. Given that the first dice shows a 2, find the probability that $Y = 2$.

d. Given that $Y = 6$, find the probability that the second dice shows a 3.

4. A fitness club has two types of membership, full and weekend.
The club has 120 members, of whom 65 are full members.
The table shows the exercise classes attended by the members.

	Kickboxing	**Yoga**
Full	36	42
Weekend	33	31

A member is chosen at random.

a. Find:

i $\mathrm{P}(\text{Weekend} \mid \text{Yoga})$

ii $\mathrm{P}(\text{Full} \mid \text{Kickboxing})$

iii $\mathrm{P}(\text{Yoga} \mid \text{Full})$

b. Decide whether taking yoga and kickboxing are mutually exclusive.
Give a reason for your answer.

5. An integer between 1 and 20 inclusive is selected at random.
Given that it is not a prime number, find the probability that it is a square number.

6. E/P A group of 140 sixth-form students are asked about their A level subjects. 82 study humanities subjects (H) and 64 study languages (L). Of those who study humanities, 36 also study languages.

a. Draw a two-way table to show this information. (2 marks)

b. One student is chosen at random. Find:

i $\mathrm{P}(L')$

ii $\mathrm{P}(H' \cap L)$

iii $\mathrm{P}(H \mid L)$

iv $\mathrm{P}(L \mid H')$

(4 marks)

c. Decide whether the events L and H are independent. Give a reason for your answer. (2 marks)

7. E/P A hotel had 540 guests during one month. Of these guests, 290 were male. 320 of the guests had the full breakfast and 160 of the guests had the continental breakfast. Of the remaining guests, 20 are female guests who did not have breakfast. 75 male guests had the continental breakfast. One guest is chosen at random. Given that:

M is the event that a guest is male

C is the event that a guest had the continental breakfast

F is the event that a guest had the full breakfast

Find:

a. $\mathrm{P}(C' \cap F')$ (1 mark)

b. $\mathrm{P}(M \mid C')$ (1 mark)

c. $\mathrm{P}(F \mid M)$ (1 mark)

d. $\mathrm{P}((C' \cap F') \mid M)$ (1 mark)

8. E/P The eating tastes of 60 cats are recorded. There are 38 adult cats, of which 11 like Furina, 17 like Purskers and the rest like Whilix. Of the kittens, 7 like Furina, 8 like Whilix and the rest like Purskers.

a. Draw a two-way table to show this situation. (2 marks)

One cat is chosen at random.

b. Find:

i $\mathrm{P}(\text{Adult cat} \mid \text{Furina})$

ii $\mathrm{P}(\text{Kitten} \mid \text{Whilix})$

iii $\mathrm{P}(\text{Purskers} \mid \text{Kitten})$

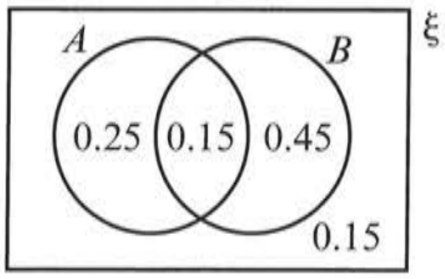
(3 marks)

c. Given that one-third of the cats who like Furina are male, two-thirds of the cats who like Purskers are male and half of the cats who like Whilix are male, find the probability that a randomly chosen cat is male. (2 marks)

2.3 Conditional probabilities in Venn diagrams

Practice Book, printed page 13.

1. The Venn diagram shows the probabilities for two events, A and B.



Find:

a. $\mathrm{P}(A \cup B)$

b. $\mathrm{P}(A \mid B)$

c. $\mathrm{P}(B' \mid A)$

d. $\mathrm{P}(B \mid A \cup B)$

Hint

You can find conditional probabilities from a Venn diagram by considering the section of the Venn diagram that corresponds to the restricted sample space. For example, to find $\mathrm{P}(A \mid B)$, just consider the circle B.

2. C and D are two events such that $\mathrm{P}(C) = 0.6$, $\mathrm{P}(D) = 0.5$ and $\mathrm{P}(C \cap D) = 0.3$.

a. Draw a Venn diagram showing the probabilities for events C and D.

b. Find:

i $\mathrm{P}(C \cup D)$

ii $\mathrm{P}(C \mid D)$

iii $\mathrm{P}(D \mid C')$

iv $\mathrm{P}(D' \mid C)$

Hint

To find $\mathrm{P}(D \mid C')$ only consider the area outside the circle C.

3. S and T are two events such that $\mathrm{P}(S) = 0.25$ and $\mathrm{P}(T) = 0.8$.
Given that S and T are independent,

a. draw a Venn diagram showing the probabilities for events S and T

b. find:

i $\mathrm{P}(S \cap T)$

ii $\mathrm{P}(S \mid T')$

iii $\mathrm{P}(T' \mid S)$

iv $\mathrm{P}(S' \cup T \mid T)$

4. A bag contains 150 counters. Some of the counters are black (B) and some of the counters are triangular (T). The bag contains 50 black counters, 75 triangular counters and 25 black triangular counters. A counter is picked from the bag at random. Find:

a. $\mathrm{P}(B' \cap T)$

b. $\mathrm{P}(T \mid B)$

c. $\mathrm{P}(B \mid T')$

d. $\mathrm{P}(B \mid B \cup T)$

5. A group of 50 students are asked whether they like burgers or curries. 36 said they liked burgers (B) and 24 said they liked curries (C), but 10 said they liked neither.
One student is chosen at random. Find:

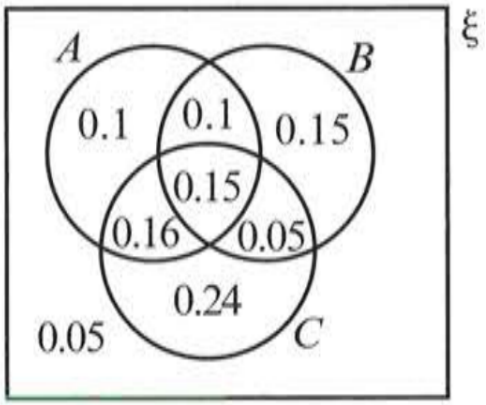
a. $\mathrm{P}(B \cap C)$

b. $\mathrm{P}(C \mid B)$

c. $\mathrm{P}(B \mid C)$

d. $\mathrm{P}(B' \mid C')$

6. The Venn diagram shows the probabilities of three events, A, B and C.



Find:

a. $\mathrm{P}(A \mid B)$

b. $\mathrm{P}(B' \mid C)$

c. $\mathrm{P}((A \cup C) \mid B')$

d. $\mathrm{P}(A \mid (B' \cap C'))$

7. E/P From the large data set, the daily total rainfalls, daily mean temperatures and daily total hours of sunshine were recorded for the 30 days of September in Hurn in 2015.
The event R is that the daily total rainfall is greater than the monthly mean.
The event S is that the daily total hours of sunshine is greater than the monthly mean.
The event T is that the daily mean temperature is greater than the monthly mean.
The table summarises the data.

Event	**Number of days**
only R	3
R and T but not S	4
only S	12
S and T but not R	4
only T	5
not R, S or T	2

a. Draw a Venn diagram to represent this information. (3 marks)

b. One day is selected at random. Given that only one of the monthly means has been exceeded, find the probability that the daily total rainfall has exceeded its monthly mean. (2 marks)

c. Determine whether events S and T are statistically independent. (3 marks)

8. E/P Three events, A, B and C, are such that A and B, A and C, and B and C are independent. $\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.1$, $\mathrm{P}(C) = 0.25$ and $\mathrm{P}(A \cap B \cap C) = 0.01$.

a. Draw a Venn diagram to show the probabilities for events A, B and C. (4 marks)

b. Find:

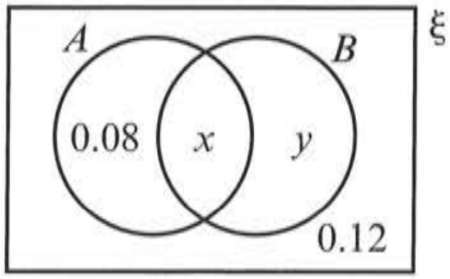
i $\mathrm{P}(A \mid C)$

ii $\mathrm{P}(B' \mid A')$

iii $\mathrm{P}(C \mid (A \cup B))$

(3 marks)

9. E/P The Venn diagram shows the probabilities for two events, A and B. Given that $\mathrm{P}(A \mid B) = \mathrm{P}(A)$, find the values of x and y. (3 marks)



2.4 Probability formulae

Practice Book, printed page 15.

1. A and B are two events.
$\mathrm{P}(A) = 0.5$, $\mathrm{P}(B) = 0.3$ and $\mathrm{P}(A \cup B) = 0.6$.
Find:

a. $\mathrm{P}(A \cap B)$

b. $\mathrm{P}(A')$

c. $\mathrm{P}(A' \cup B)$

d. $\mathrm{P}(A \cup B')$

Hint

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

2. X and Y are two events.
$\mathrm{P}(X) = 0.45$, $\mathrm{P}(Y) = 0.6$ and $\mathrm{P}(X \cup Y) = 0.75$.

a. Use the probability addition formula to find $\mathrm{P}(X \cap Y)$.

b. Using a Venn diagram, or otherwise, find:

i $\mathrm{P}(X' \cap Y')$

ii $\mathrm{P}(Y \mid X)$

iii $\mathrm{P}(Y \mid X')$

c. State, with a reason, whether events X and Y are statistically independent.

3. A and B are two events.
$\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.7$ and $\mathrm{P}(A \mid B) = 0.5$.
Find:

a. $\mathrm{P}(A \cap B)$

b. $\mathrm{P}(B \mid A)$

c. $\mathrm{P}(A \cup B)$

Hint

$$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(B \cap A)}{\mathrm{P}(A)}$$
$$\mathrm{P}(B \cap A) = \mathrm{P}(B \mid A) \times \mathrm{P}(A)$$

4. P E and F are two events where $\mathrm{P}(E) = x$, $\mathrm{P}(F) = 1.5x$, $\mathrm{P}(E \cap F) = 0.3$ and $\mathrm{P}(E' \cap F') = 0.05$.

a. Use this information to write $\mathrm{P}(E \cup F)$ in terms of x.

b. Hence show that $x = 0.5$.

c. Find:

i $\mathrm{P}(F)$

ii $\mathrm{P}(E \cap F')$

iii $\mathrm{P}(E \mid F')$

5. Some students were asked whether their families owned pets. 40% said they owned a dog and 25% said they owned a cat. Half of the students said they owned either a dog or a cat or both. Find the probability that a randomly chosen student owns a cat but not a dog.

6. 60% of visitors at an outdoor activity park go abseiling and 30% go canoeing. Given that a visitor goes canoeing, there is a 50% chance that they will go abseiling. A visitor is chosen at random.

a. Find the probability that they:

i went canoeing but not abseiling

ii didn't go abseiling or canoeing.

b. Given that the visitor chosen went abseiling, find the probability that they went canoeing.

7. C and D are two events where $\mathrm{P}(C \mid D) = 0.2$, $\mathrm{P}(C \mid D') = 0.6$ and $\mathrm{P}(D) = 0.5$. Find:

a. $\mathrm{P}(C \cap D)$

b. $\mathrm{P}(C \cap D')$

c. $\mathrm{P}(C)$

d. $\mathrm{P}(D \mid C)$

e. $\mathrm{P}(D' \mid C)$

f. $\mathrm{P}(D' \mid C')$

8. E/P Given that $\mathrm{P}(A) = 0.5$, $\mathrm{P}(B) = 0.35$ and $\mathrm{P}(A \cap B) = 0.24$, find:

a. $\mathrm{P}(A \cup B)$ (2 marks)

b. $\mathrm{P}(A' \mid B)$ (2 marks)

The events B and C are mutually exclusive and the events A and C are independent.
$\mathrm{P}(A \cap C) = 0.14$

c. Find $\mathrm{P}(C)$. (2 marks)

d. Draw a Venn diagram to illustrate the events A, B and C, showing the probabilities for each region. (4 marks)

e. Find $\mathrm{P}((A \cup C')')$. (2 marks)

9. E/P Three events A, B and C are such that $\mathrm{P}(A) = 0.5$, $\mathrm{P}(B) = 0.4$, $\mathrm{P}(C) = 0.2$ and $\mathrm{P}(A \cap B) = 0.1$.
Given that A and C are mutually exclusive and that B and C are independent, find:

a. $\mathrm{P}(B \cap C)$ (1 mark)

b. $\mathrm{P}(B \mid A)$ (1 mark)

c. $\mathrm{P}(A' \mid B)$ (1 mark)

d. $\mathrm{P}((B \cap C) \mid A')$ (1 mark)

10. E/P The probability that Zahira gets home from work before 6 pm is 0.4.
The probability that she goes to the gym in the evening is 0.6.
The probability that she gets home from work after 6 pm and does not go to the gym is 0.35.
Let H represent the event that Zahira gets home after 6 pm and G represent the event that she goes to the gym in the evening.
Find:

a. $\mathrm{P}(G \cup H)$ (2 marks)

b. $\mathrm{P}(G \cap H')$ (2 marks)

c. $\mathrm{P}(H \mid G)$ (3 marks)

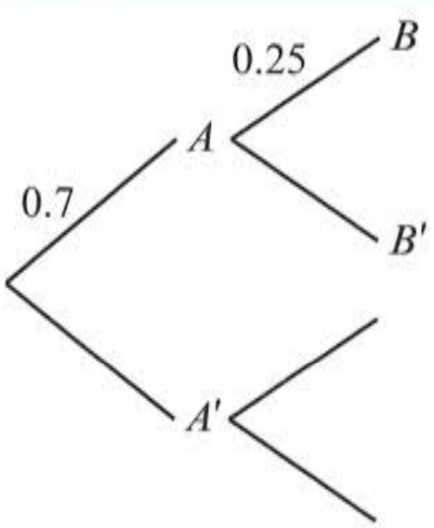
d. State, with a reason, whether or not events G and H are independent. (2 marks)

2.5 Tree diagrams

Practice Book, printed page 17.

1. A and B are two events such that $\mathrm{P}(B \mid A) = 0.25$, $\mathrm{P}(B \mid A') = 0.6$ and $\mathrm{P}(A) = 0.7$.

a. Copy and complete the tree diagram to represent this information.



b. Find:

i $\mathrm{P}(A \cap B)$

ii $\mathrm{P}(A' \cap B')$

iii $\mathrm{P}(B' \mid A)$

iv $\mathrm{P}(A \mid B)$

2. A packet of 32 bulbs contains 18 tulips and 14 daffodils. A gardener chooses a bulb at random; the type of bulb is recorded and the bulb is not replaced. A second bulb is chosen and the type recorded.

a. Draw a tree diagram to represent this information.

Find the probability the gardener chooses:

b. two tulip bulbs

c. one tulip and one daffodil

d. two daffodils given that he chooses at least one daffodil.

Hint

Since the bulbs are removed and not replaced, the total number of bulbs is reduced to 31 for the second pick.

3. P Max has two bags. Bag A contains the same number of black counters and white counters. Bag B contains twice as many black counters as white counters.
Max chooses one of the bags at random and chooses a counter.

a. Find the probability he chooses a black counter.

b. Given that he chooses a white counter, find the probability that he chose from bag A.

4. P Dev goes to gym every morning before work. He always uses either the treadmill or the exercise bike. If he uses the treadmill one day, the probability he uses the treadmill the next day is 0.5. If he uses the exercise bike one day, the probability he uses the exercise bike the next day is 0.6.
Dev uses the treadmill on Monday. Given that he also uses the treadmill on Thursday, find the probability that he used the exercise bike on Tuesday.

5. E/P A fish tank contains 5 rainbowfish and 6 goldfish. A fish is selected at random from the tank and its type is recorded. The fish is moved to another tank. A second fish is selected at random and its type is recorded.

a. Draw a tree diagram to represent the information. (3 marks)

Find the probability that:

b. the second fish selected is a rainbowfish (2 marks)

c. both fish selected are rainbowfish, given that the second fish is a rainbowfish. (2 marks)

6. E/P During a football season, players A, B and C take all of the penalties for a team. Player A takes 35% of the penalties, player B takes 50% and the rest are taken by player C. Players A, B and C score penalties with probabilities 0.7, 0.8 and 0.6 respectively.

a. Draw a tree diagram to represent this information. (3 marks)

b. Find the probability that a randomly selected penalty is:

i taken by player B and missed

ii scored.

(5 marks)

c. Given that a randomly selected penalty is missed, find the probability that it was taken by player C. (3 marks)

7. E/P A factory produces microchips and 95% of the microchips have no defects.
A quality control test is applied to help determine whether or not a microchip is defective.
If a microchip has no defects, it passes the quality control test with probability 0.95.
If a microchip has defects, it passes the quality control with probability 0.08.

a. Draw a tree diagram to represent this information. (3 marks)

b. A microchip is selected at random and tested for defects. Find the probability that it passes the quality control test. (3 marks)

c. A microchip is selected at random and tested for defects. Given that it fails the quality control test, find the probability that the microchip has no defects. (2 marks)

d. Comment on the effectiveness of the quality control test. (1 mark)

8. E/P Candidates taking a professional exam either pass, fail or are retested, with probabilities 0.1, 0.4 and 0.5 respectively. When a candidate is retested for the first time, the three possible outcomes and their probabilities remain the same as for the original test. When a candidate is retested for the second time there are just two possible outcomes: pass or fail, with probabilities 0.2 and 0.8 respectively.

a. Draw a probability tree diagram to illustrate the outcomes. (3 marks)

b. Find the probability that a randomly selected candidate passes. (2 marks)

c. Find the probability that a randomly selected candidate is retested at least once, given that this candidate passes. (3 marks)

Problem solving: Set A

Bronze

Bag A contains 7 black counters and 9 white counters. Bag B contains 5 black counters and 6 white counters. A counter is chosen at random from bag A and its colour is recorded. The counter is then added to bag B. A counter is then chosen at random from bag B and its colour is recorded.

a. Draw a tree diagram to represent this information. (3 marks)

Find the probability of choosing:

b. a black counter from bag B (2 marks)

c. a white counter from bag A, given that a white counter is chosen from bag B. (2 marks)

Silver

A bag contains 11 red dice and 10 blue dice. The red dice are all six-sided and fair. The blue dice are also six-sided but are biased so that they are three times as likely to land on 4 as on any other outcome. A dice is chosen at random from the bag and thrown. Give that the score on the dice is 4, find the probability that the dice is blue. (5 marks)

Gold

A bag contains red and blue counters. There are three times as many blue counters as red. Each counter is numbered 1, 2, 3, 4 or 5. If a red counter is chosen at random, the number shown on the counter, X, has probability distribution

$$\mathrm{P}(X = x) = \begin{cases} \frac{x}{15} & x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

If a blue counter is chosen at random, the number shown on the counter, Y, has probability distribution

$$\mathrm{P}(Y = y) = \begin{cases} \frac{4-y}{15} & y = 1, 2, 3 \\ a & y = 4 \\ b & y = 5 \\ 0 & \text{otherwise} \end{cases}$$

where a and b are constants. A counter is chosen at random from the bag and the number shown on the counter is recorded.

a. Find $\mathrm{P}(1)$. (2 marks)

b. Given that $\mathrm{P}(\text{Blue} \mid 4) = \frac{27}{43}$, find the values of a and b. (4 marks)

Problem solving: Set B

Bronze

Yvonne and Zoe both work for the same sales department. Y is the event that Yvonne does not meet her weekly sales target. Z is the event that Zoe does not meet her weekly sales target. $\mathrm{P}(Y) = 0.15$, $\mathrm{P}(Y \cap Z) = 0.05$ and $\mathrm{P}(Y' \cap Z') = 0.75$.

a. For a randomly selected week, find the probability that:

i at least one of Yvonne or Zoe do not meet their weekly sales target

ii Zoe does not meet her weekly sales target.

(3 marks)

b. Given that Zoe does not meet her weekly sales target, find the probability that Yvonne does not meet her target. (3 marks)

The sales manager suspects that Yvonne not meeting her sales target and Zoe not meeting her sales target are linked in some way.

c. Determine whether Y and Z are statistically independent. (2 marks)

d. Comment on the manager's suspicion in the light of your calculation in part **d**. (1 mark)

Silver

The events A and B are independent and such that $\mathrm{P}(A) = 3\mathrm{P}(B)$ and $\mathrm{P}(A \cup B) = \frac{7}{12}$

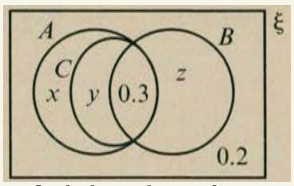
a. Show that $\mathrm{P}(B) = \frac{1}{6}$ (5 marks)

b. Find $\mathrm{P}(A \cap B)$. (3 marks)

c. Find $\mathrm{P}(B \mid A')$. (2 marks)

Gold

The events A, B and C are shown on this Venn diagram.



Given that $\mathrm{P}(A \mid B) = 0.6$ and $\mathrm{P}(C \mid A) = 0.75$, find the values of x, y and z. (7 marks)

Selected exam question bank questions

Question 1

A choir is made up of 18 women and 15 men. Of these, 10 of the women and 6 of the men can read music.

a. Draw a two-way table to show this information. (2 marks)

b. One member is chosen at random. Find:

i P(Reads music)

ii P(Can't read music | Female)

iii P(Male | Reads music)

(3 marks)

Question 5

Two events A and B are such that $\mathrm{P}(A \mid B) = \mathrm{P}(A) = \mathrm{P}(B) = \mathrm{P}(B \mid A) = \mathrm{P}(A' \cap B')$.

Find $\mathrm{P}(A)$. (4 marks)

Question 9

Abi and Brianna are both taking their driving tests.

A is the event that Abi passes her driving test.

B is the event that Brianna passes her driving test.

$\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.3$ and $\mathrm{P}(A \cup B) = 0.58$.

Find the probability that:

a. both Abi and Brianna pass their driving tests (1 mark)

b. Brianna fails her test given that Abi passes (2 marks)

c. Abi passes her test given that Brianna passes (2 marks)

d. State whether the events A and B are statistically independent. You must give a reason for your answer. (1 mark)

Question 12

Members of a film club watch comedies (C), sci-fi (S) and foreign language films (F).

$\mathrm{P}(C) = 0.85$, $\mathrm{P}(S) = 0.6$ and $\mathrm{P}(F) = 0.3$.

No one watches both sci-fi films and foreign language films and $\mathrm{P}(C \cap S) = 0.45$.

a. Draw a Venn diagram to illustrate these probabilities. (4 marks)

b. Find:

i $\mathrm{P}(F \cup S)$

ii $\mathrm{P}((C \cap S) \cup (C \cap F))$

(2 marks)

c. Are the events F' and S independent? You must justify your answer. (2 marks)

Question 16

A factory produces metal components. Out of 250 components, 15 are known to be defective. A machine correctly identifies 12 of the defective components as defective. It incorrectly identifies 4 components without defects as defective.

The event D is 'component is defective' and the event M is 'machine identifies the component as defective'.

a. Draw a Venn diagram to represent this situation. (3 marks)

b. Calculate $\mathrm{P}(M \mid D)$. (2 marks)

c. Calculate $\mathrm{P}(D' \mid M)$. (2 marks)

d. With reference to your answer to part **b**, comment on the usefulness of the machine. (1 mark)

Question 18

Serving against his regular opponent, a tennis player has a 70% chance of getting his first serve in. If his first serve is in, he then has a 60% chance of winning the point, but if his first serve is not in, he only has a 30% chance of winning the point.

a. Represent this information on a tree diagram. (2 marks)

For a point on which this player served to his regular opponent, find the probability that:

b. he won the point (2 marks)

c. his first serve went in given that he won the point (2 marks)

d. his first serve didn't go in given that he lost the point. (3 marks)

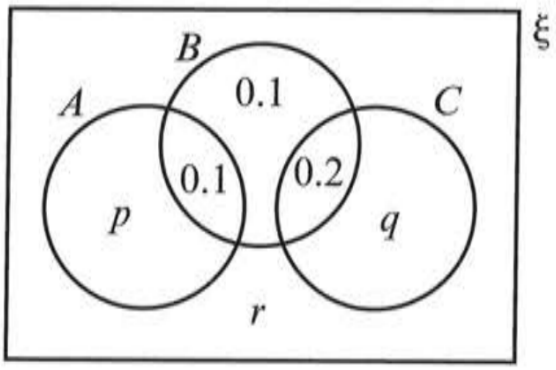
The tennis player decides to try a new tennis racket, and tests it by serving 100 times.

He finds that his first serve goes in on 62 occasions.

e. Test, at the 5% level of significance, whether the new racket has changed his probability of getting his first serve in. State any assumptions you have made. (5 marks)

Question 20

The Venn diagram shows the probabilities of three events A, B and C.



The events A and B are independent.

a. Find the value of p. (3 marks)

b. Given that $\mathrm{P}(B \mid C) = \frac{4}{11}$, find the value of q and the value of r. (4 marks)

c. Find $\mathrm{P}(A \cup C \mid B)$. (2 marks)

Answers

2.1 Set notation

1.

a. $A' \cap B$

b. $(A \cup B)'$

c. $(A \cap B') \cup (A' \cap B)$

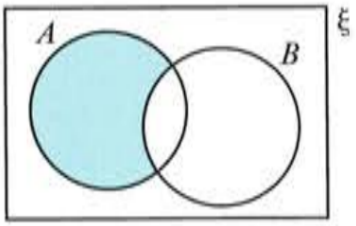
d. $(A \cup B)'$

e. $(A \cap B) \cup (B \cap C) \cup (A \cap C)$

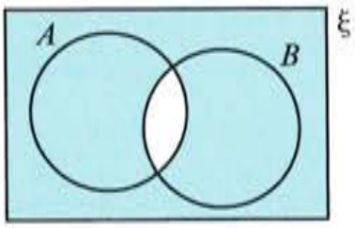
f. $A \cap C \cap B'$

2.

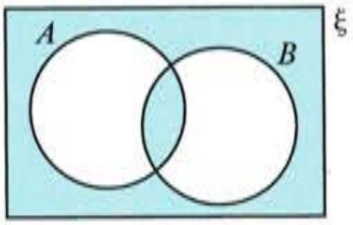
a.



b.

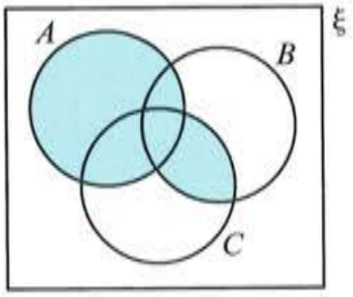


c.

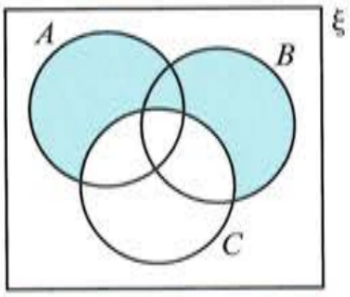


3.

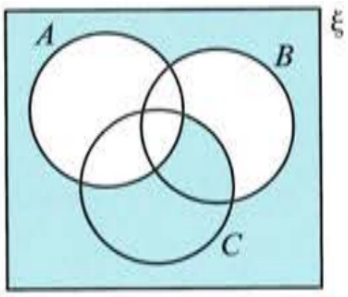
a.



b.



c.



4.

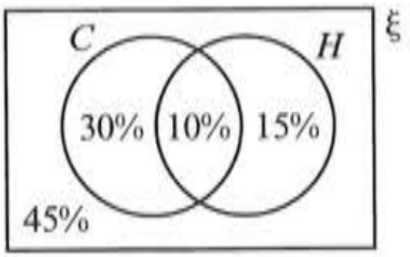
a. $\frac{3}{13}$
b. $\frac{1}{4}$
c. $\frac{3}{52}$
d. $\frac{11}{26}$
e. $\frac{10}{13}$
f. $\frac{9}{52}$

5.

a. 0.5
b. 0.6
c. 0.1
d. 0.5

6.

a.

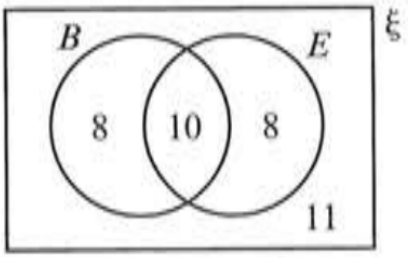


b.

i 0.1
ii 0.3
iii 0.15

7.

a.

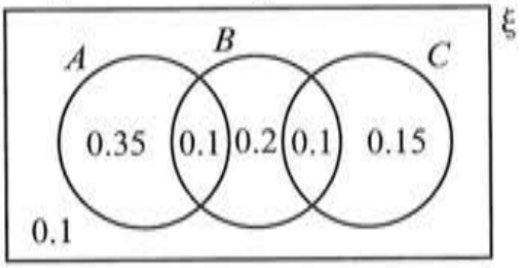


b.

i $\frac{26}{37}$
ii $\frac{8}{37}$
iii $\frac{27}{37}$

8.

a.



b.

i $\mathrm{P}(A \cap C) = 0$
ii $\mathrm{P}(B) \times \mathrm{P}(C) = 0.4 \times 0.25 = 0.1 = \mathrm{P}(B \cap C)$

c.

i 0.3
ii 0.8
iii 0.2

2.2 Conditional probability

1.

a. $\frac{21}{40}$

b. $\frac{4}{7}$

c. $\frac{9}{20}$

d. $\frac{11}{19}$

2.

a.

		Yellow spinner			
		1	2	3	4
Green spinner	1	1	2	3	4
	2	2	4	6	8
	3	3	6	9	12
	4	4	8	12	16

b.

i $\frac{1}{8}$

ii $\frac{1}{4}$

iii $\frac{1}{3}$

3.

a.

		Dice 1					
		1	2	3	4	5	6
Dice 2	1	1	2	3	4	5	6
	2	2	2	3	4	5	6
	3	3	3	3	4	5	6
	4	4	4	4	4	5	6
	5	5	5	5	5	5	6
	6	6	6	6	6	6	6

b. $\frac{7}{36}$

c. $\frac{1}{3}$

d. $\frac{1}{11}$

4.

a.

i $\frac{31}{73}$

ii $\frac{12}{23}$

iii $\frac{42}{78}$

b. Not mutually exclusive. The total number of people in all of the classes is 142. So there are 22 members that go to both classes.

5. $\frac{1}{3}$

6.

a.

	H	H'	**Total**
L	36	28	64
L'	46	30	76
Total	82	58	140

b.

i $\frac{19}{35}$
ii $\frac{1}{5}$
iii $\frac{9}{16}$
iv $\frac{14}{29}$

c. $\mathrm{P}(L) \times \mathrm{P}(H) = \frac{16}{35} \times \frac{41}{70} = \frac{328}{1225} \neq \frac{9}{35} = \mathrm{P}(L \cap H)$. Therefore L and H are not independent.

7.

a. $\frac{1}{9}$
b. $\frac{43}{76}$
c. $\frac{35}{58}$
d. $\frac{4}{29}$

8.

a.

	Furina	**Purskers**	**Whilix**	**Total**
Adult cats	11	17	10	38
Kittens	7	7	8	22
Total	18	24	18	60

b.

i $\frac{11}{18}$
ii $\frac{4}{9}$
iii $\frac{7}{22}$

c. $\frac{31}{60}$

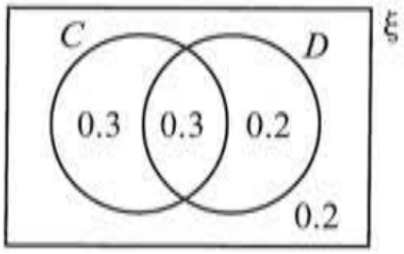
2.3 Conditional probabilities in Venn diagrams

1.

a. 0.85
b. 0.25
c. 0.625
d. $\frac{12}{17}$

2.

a.

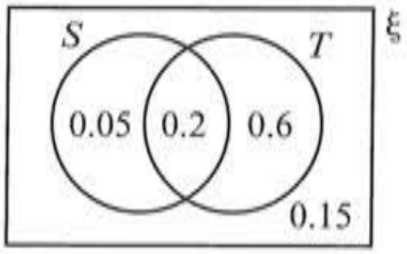


b.

i 0.8
ii 0.6
iii 0.5
iv 0.5

3.

a.



b.

i 0.2
ii 0.25
iii 0.8
iv 1

4.

a. $\frac{1}{3}$
b. 0.5
c. $\frac{1}{3}$
d. 0.5

5.

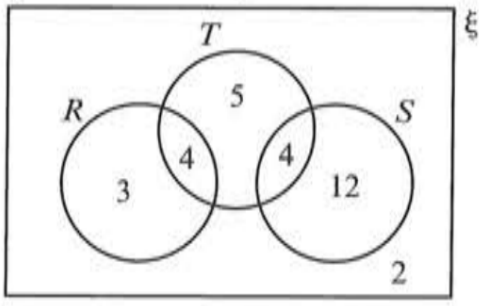
a. 0.4
b. $\frac{5}{9}$
c. $\frac{5}{6}$
d. $\frac{5}{13}$

6.

a. $\frac{5}{9}$
b. $\frac{2}{3}$
c. $\frac{10}{11}$
d. $\frac{2}{3}$

7.

a.

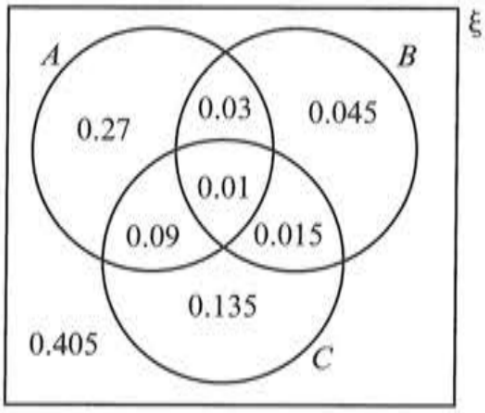


b. $\frac{3}{20}$

c. Not independent

8.

a.



b.

i 0.4

ii 0.9

iii 0.25

9. $x = 0.32$, $y = 0.48$

2.4 Probability formulae

1.

a. 0.2

b. 0.5

c. 0.7

d. 0.9

2.

a. 0.3

b.

i 0.25

ii $\frac{2}{3}$

iii $\frac{6}{11}$

c. No. For example, because $\mathrm{P}(Y \mid X) \neq \mathrm{P}(Y)$

3.

a. 0.35

b. 0.875

c. 0.75

4.

a. $\mathrm{P}(E \cup F) = 2.5x - 0.3$

b. $\mathrm{P}(E \cup F) + \mathrm{P}(E' \cap F') = 1$
$2.5x - 0.3 + 0.05 = 1$
$\Rightarrow 2.5x = 1.25$
$\Rightarrow x = 0.5$

c.

i 0.75

ii 0.2

iii 0.8

5. 0.1

6.

a.

i 0.15

ii 0.25

b. 0.25

7.

a. 0.1

b. 0.3

c. 0.4

d. 0.25

e. 0.75

f. $\frac{1}{3}$

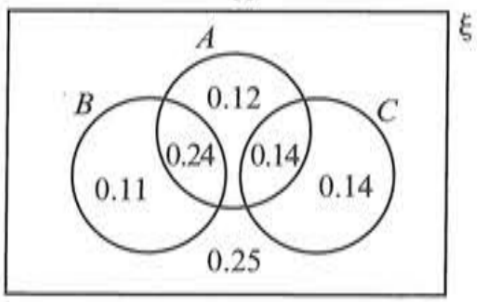
8.

a. 0.61

b. $\frac{11}{35}$

c. 0.28

d.



e. 0.14

9.

a. 0.08

b. 0.2

c. 0.75

d. 0.16

10.

a. 0.95

b. 0.35

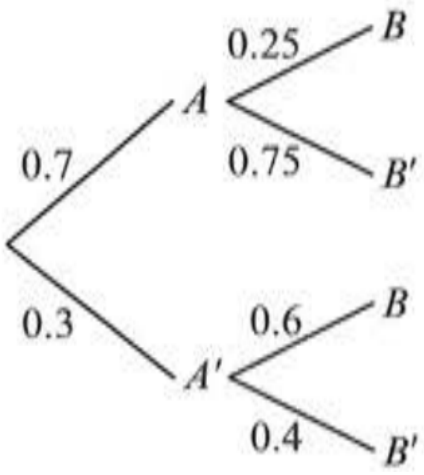
c. $\frac{5}{12}$

d. Not independent, since $\mathrm{P}(H \mid G) \neq \mathrm{P}(H)$

2.5 Tree diagrams

1.

a.



b.

i 0.175

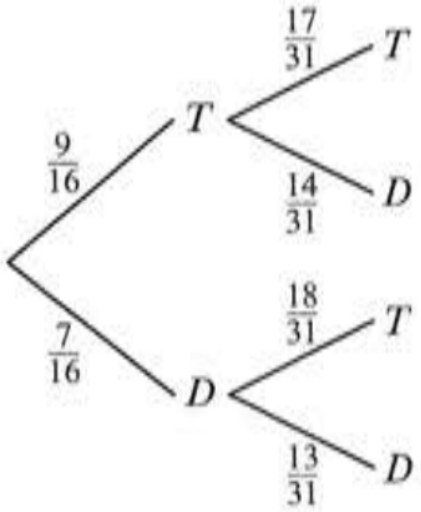
ii 0.12

iii 0.75

iv $\frac{35}{71}$

2.

a.



b. $\frac{153}{496}$

c. $\frac{63}{124}$

d. $\frac{13}{49}$

3.

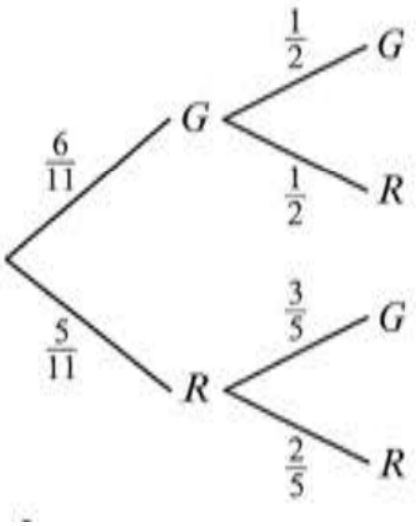
a. $\frac{7}{12}$

b. 0.6

4. $\frac{44}{89}$

5.

a.

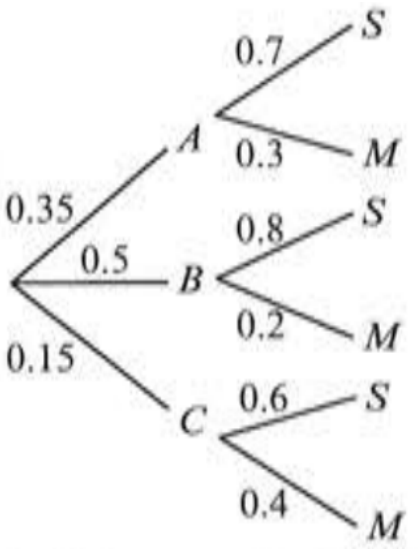


b. $\frac{5}{11}$

c. 0.4

6.

a.



b.

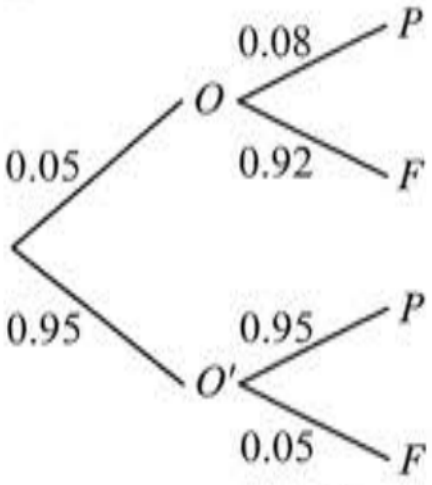
i 0.1

ii 0.735

c. $\frac{12}{53}$

7.

a.



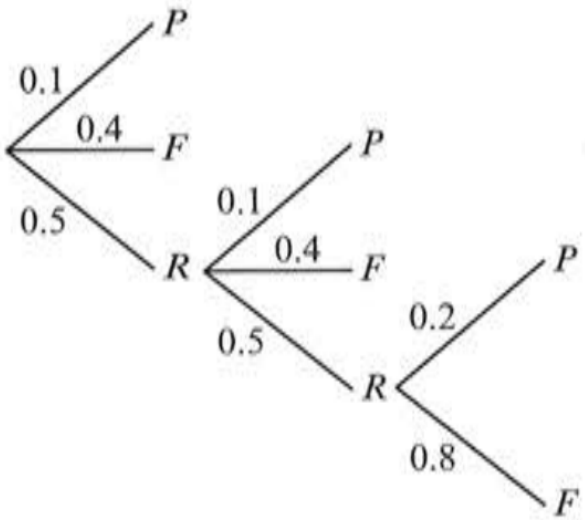
b. 0.9065

c. $\frac{95}{187}$

d. 49% of the microchips that fail the test will not have defects. Therefore the test is not effective.

8.

a.



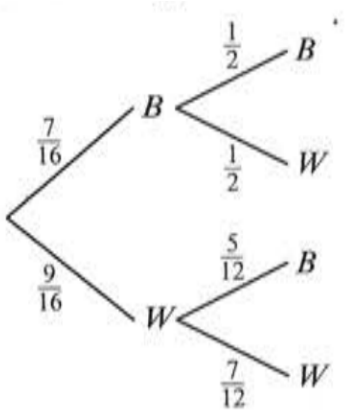
b. 0.2

c. 0.5

Problem solving: Set A

Bronze

a.



b. $\frac{29}{64}$
c. $\frac{3}{5}$

Silver

$\frac{45}{67}$

Gold

a. $\frac{1}{6}$
b. $a = 0.15$, $b = 0.45$

Problem solving: Set B

Bronze

a.

i 0.25

ii 0.15

b. $\frac{1}{3}$
c. Not independent: $\mathrm{P}(Y \mid Z) = \frac{1}{3} \neq \mathrm{P}(Y) = 0.15$.
d. If Zoe doesn't meet her target, Yvonne is more likely to miss hers.

Silver

a. $\mathrm{P}(B) = x$, $\mathrm{P}(A) = 3x$, $\mathrm{P}(A \cap B) = 3x^2$
$3x + x - 3x^2 = \frac{7}{12}$
$\Rightarrow 3x^2 - 4x + \frac{7}{12} = 0$
$\Rightarrow x = \frac{1}{6}$
b. $\frac{1}{12}$
c. $\frac{1}{6}$

Gold

$x = 0.15$, $y = 0.15$ and $z = 0.2$

Selected exam question bank answers

Question 1

a.

	Reads music	Can't read music	Total
Female	10	8	18
Male	6	9	15
Total	16	17	33

b.

i $\frac{16}{33}$

ii $\frac{4}{9}$

iii $\frac{3}{8}$

Question 5 $\frac{3-\sqrt{5}}{2}$

Question 9

a. 0.12

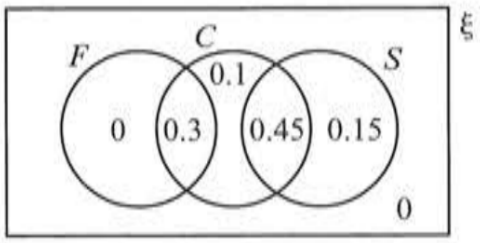
b. 0.7

c. 0.4

d. $\mathrm{P}(A) = \mathrm{P}(A \mid B)$, therefore independent.

Question 12

a.



b.

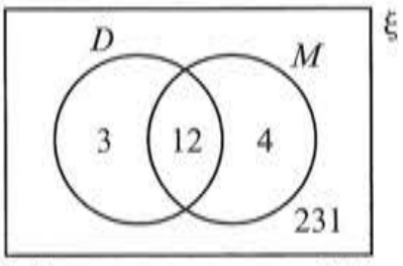
i 0.9

ii 0.75

c. No, $\mathrm{P}(F' \cap S) = 0.6 \neq \mathrm{P}(F') \times \mathrm{P}(S) = 0.7 \times 0.6 = 0.42$

Question 16

a.



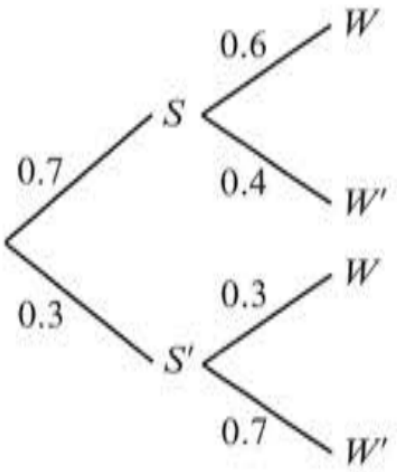
b. 0.8

c. 0.25

d. 20% of the defective components are not identified by the machine and 25% of the components identified as defective are not so test is not very effective.

Question 18

a.



b. 0.51

c. $\frac{14}{17}$

d. $\frac{3}{7}$

e. Assuming that each serve is independent and the probability of success is constant.
$H_0 : p = 0.7$, $H_1 : p \neq 0.7$. $P(X \leq 62) = 0.053 > 0.025$.
There is insufficient evidence to reject H_0. There is no evidence that proportion of first serves has changed.

Question 20

a. 0.15

b. $q = 0.35$, $r = 0.1$

c. 0.75